

Keisuke

Tradeoffs / philosophy voice

How to use this: not a verbatim script — nobody is expected to read from it word-for-word, and realistically no one will stick to a script anyway. Study it well enough to explain each point in your own words, improvise if you're cut off, and answer a follow-up without freezing. If a question lands outside your section, give the short version and hand off by name — every presenter's script and the underlying `FIELD-NOTES.md` are in the same GitHub repo.

You're the tradeoffs/philosophy voice — constraints, ethics, and the harder judgment calls sit in your slides.

Slide 5 — More constraints we have to design around.

1. Muravah has **no annual drought/flood prediction**, so they sync planting to the monsoon and specifically seek out farmers who already have water access — a real infrastructure gap.
2. The farmer population is aging (~50 average) with no government training pipeline; Muravah's own workaround is recruiting students from BU Guinubatan.
3. Hard constraint: **no synthetic fertilizers or pesticides**, biopesticides/biofertilizers only — this rules out entire categories of "obvious" agri-tech fixes before we even start.
4. Muravah is explicitly skeptical of AI and robotics — quote it directly: "not there yet, other problems to solve," and "AI is biased according to the natives." Frame this as a values position, not just a budget constraint — it's why every solution later in the deck is deliberately low-tech and human-in-the-loop.

Slide 11 — Vertical panels waste potential output.

Short, standalone point — don't over-explain it, let it land. Mounting solar panels vertically is not the efficient way to collect sunlight; a wall-mounted panel captures noticeably less irradiance than an optimally tilted one (per Masa's chart: about 65% vs. 100%). Be upfront that this is **our own engineering observation, not something a source told us in an interview** — a design point worth carrying into any agrivoltaic or solar-irrigation proposal built on the Zambales precedent. The image on this slide is a placeholder for a comparison photo not yet in the field notes.

Slide 14 — Solutions we probed and gathered from other farms.

This is the payoff slide for "adapt, don't invent." Of 12 candidate pest/disease solutions brainstormed, only **6 survived independent fact-checking** against the actual interview data — say that filtering process out loud, it's a credibility point. Three groups:

- *Monitoring & routine discipline (3)*: a cacao-adapted 15-day treatment calendar borrowed from the taro project's proven schedule; a field-officer symptom-logging checklist; a centralized compliance-and-bio-input ledger.
- *Biological & cultural control (2)*: an herbal foliar/pod spray adapted from the rice/dairy farm's practice; an amino-acid/molasses bio-concoction on the taro project's 15-day cycle.
- *Post-harvest handling (1)*: rolling phytosanitation rounds every 7–14 days.

Land the recommendation clearly: for Tatae specifically, start with the **15-day calendar plus the herbal spray** — cheapest to start, no new equipment, fits his limited labor capacity given his health constraints. The ledger and the amino-acid spray follow once the calendar habit is established.

Slide 16 — The investor gap.

Open with Tatae's own quote: he told us directly that "a pilot study and investors would be needed." Three concrete ways to help him find them:

1. **Point to the regional funding precedent** — the Zambales solar farm secured ₱10 million for a comparable-scale agri-infrastructure pilot, concrete proof that regional agri-tech investment appetite exists, and a benchmark to pitch against.
2. **Package the pilot inside Muravah's existing study** — fold it into the 500–1,000-farmer production-cycle study already underway, so investors get an established NGO's track record and scale behind it, not a lone farmer's ask.
3. **Lead with the organic price-premium ROI story** — frame the ask as bridge capital toward organic/export certification, where a projected 4× yield recovery plus a price premium is a real return case, not just a request for aid.

This is your tradeoffs lane — feel free to speak to the organic-vs-export-vs-income tension if asked.